

Intro to AI for Medicine

Lec 10: AI in the Clinic

Last Time

- Explored infrastructure available which is used to create, accelerate and deploy AI
 - GPU vs CPU
 - AI Data Centers
 - Python packages like PyTorch, TensorFlow
- Discovered ways to collaboratively train model weights
 - Federated Learning
 - Pretraining
 - Public Datasets

Today

- How do we bring everything we know into the clinic and make strong decisions that benefit patients?
- Practice thinking about ways existing AI interventions work
- How do you build a team of experts to go from idea to working application?
- What does the FDA approval process for medical devices look like?

Posing a question

1. Clinical question to answer
2. Technical problem statement
 1. Regression vs classification
 2. Supervised vs semi-supervised vs unsupervised
3. Other technical details
 1. Loss function
 2. Training regime
 3. Network architecture
4. Implementation, coding and deployment



Tips to remember

- Often many ways to approach a problem – many right answers
- 90% of the work is gathering the data
- Something working poorly is better than nothing (just make it exist)

AI Menu

	AI Tasks		
Name	Regression	Classification	Multi-class
Definition	Predicting a number	Predicting which group of two	Predicting which group of many
Examples	Hosp Stay Duration	Stroke Detection	LLMs
Activation	ReLU	Sigmoid/Tanh	Softmax
Loss Fn	Mean squared error	Log loss	Multiclass log loss
	Training Regimes		
Type	Supervised	Self-Supervised	Unsupervised
Definition	We tell model what outputs should be	Block part of data and try to guess it	Choose hparams and let model find trends

AI Menu

Special Data Types

Time Series	Imagin g	Audio
Text	Video	

Natural Language Processing

Text Classification

Token Classification

Table Question Answering

Question Answering

Zero-Shot Classification

Translation

Summarization

Feature Extraction

Text Generation

Fill-Mask

Sentence Similarity

Text Ranking

Audio

Text-to-Speech

Text-to-Audio

Automatic Speech Recognition

Audio-to-Audio

Audio Classification

Voice Activity Detection

Tabular

Tabular Classification

Tabular Regression

Time Series Forecasting

Multimodal

Audio-Text-to-Text

Image-Text-to-Text

Visual Question Answering

Document Question Answering

Video-Text-to-Text

Visual Document Retrieval

Any-to-Any

Computer Vision

Depth Estimation

Image Classification

Object Detection

Image Segmentation

Text-to-Image

Image-to-Text

Image-to-Image

Image-to-Video

Unconditional Image Generation

Video Classification

Text-to-Video

Zero-Shot Image Classification

Mask Generation

Zero-Shot Object Detection

Text-to-3D

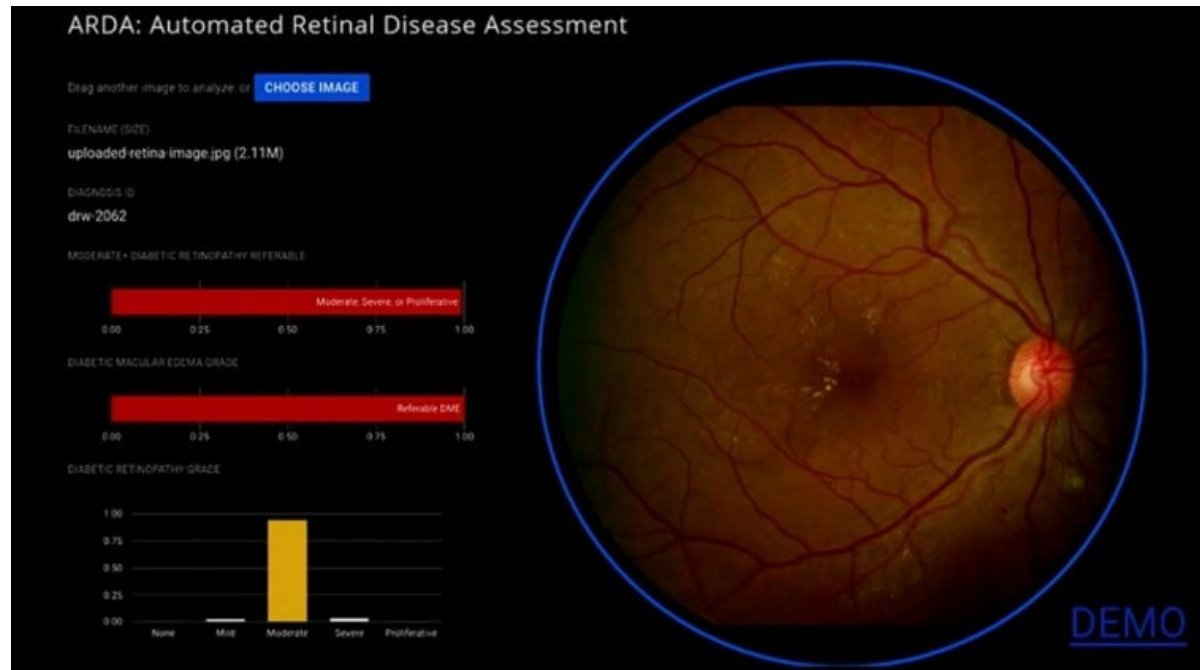
Image-to-3D

Image Feature Extraction

Keypoint Detection

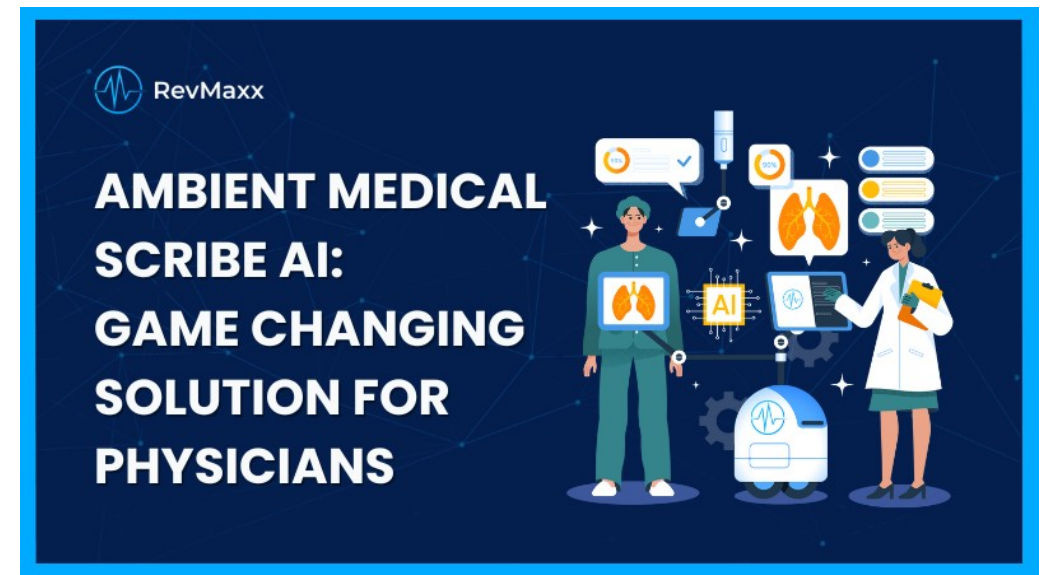
Video-to-Video

Medical AI On the Market

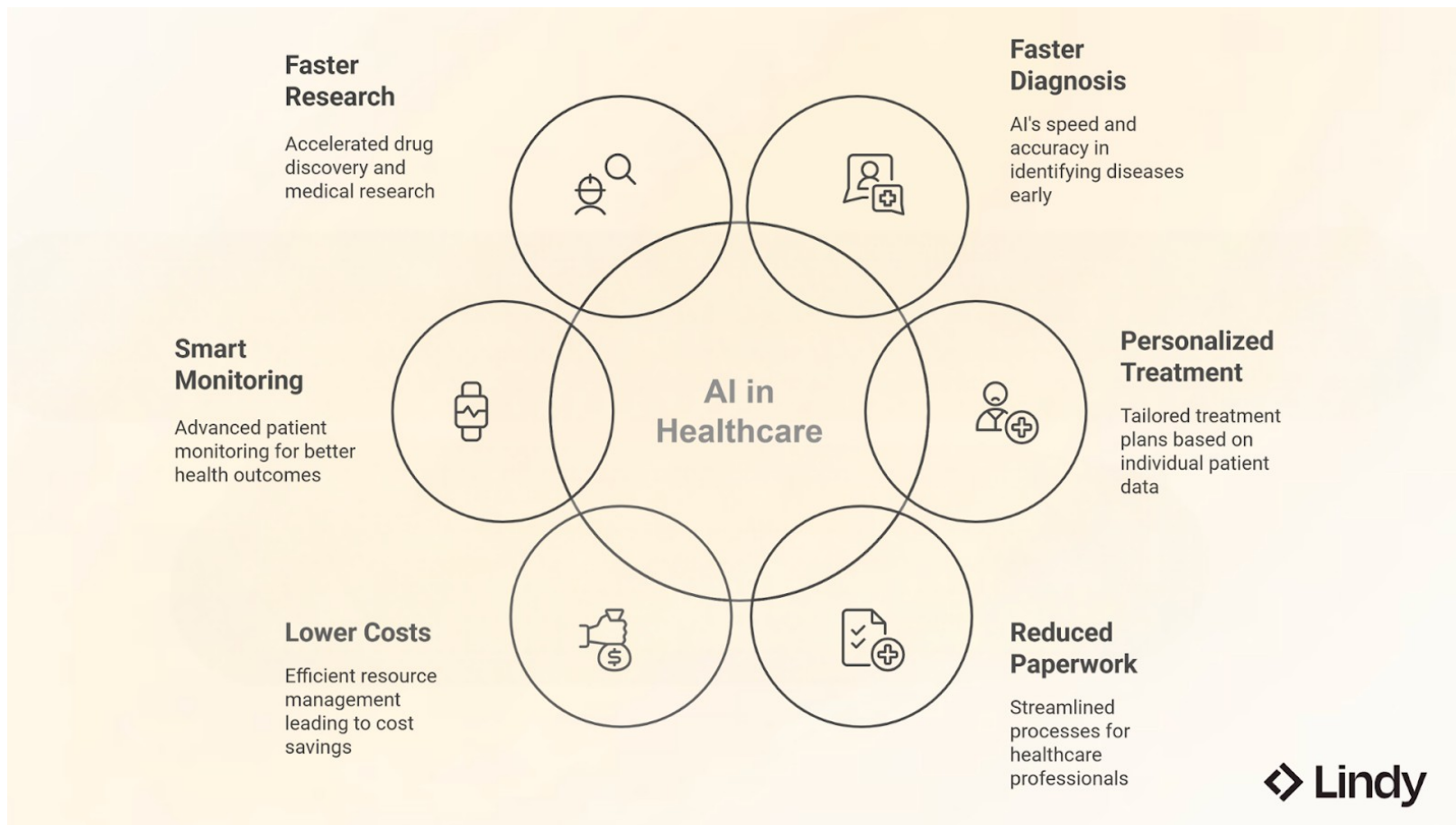


What have you guys heard of that's out there?

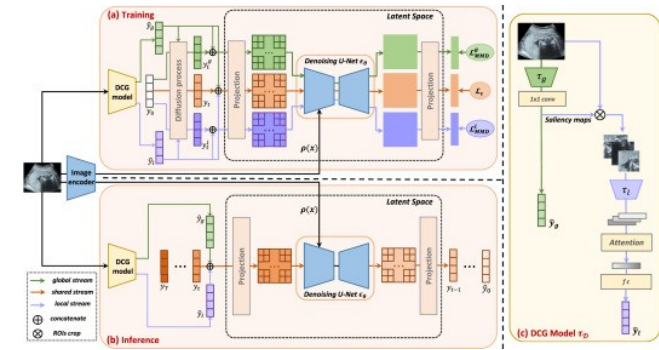
Can we use what we know to make guesses as to how these work?



Areas to Consider



Building a team



- As the clinician, your role is to pose a clinical question
 - Great investigators will keep the roles of other team members in mind
 - For example, an idea that has a certain novel network in mind is how you execute a cutting-edge idea and get engineers excited
- Clinical pain point + feasible implementation = novel success

Building a team

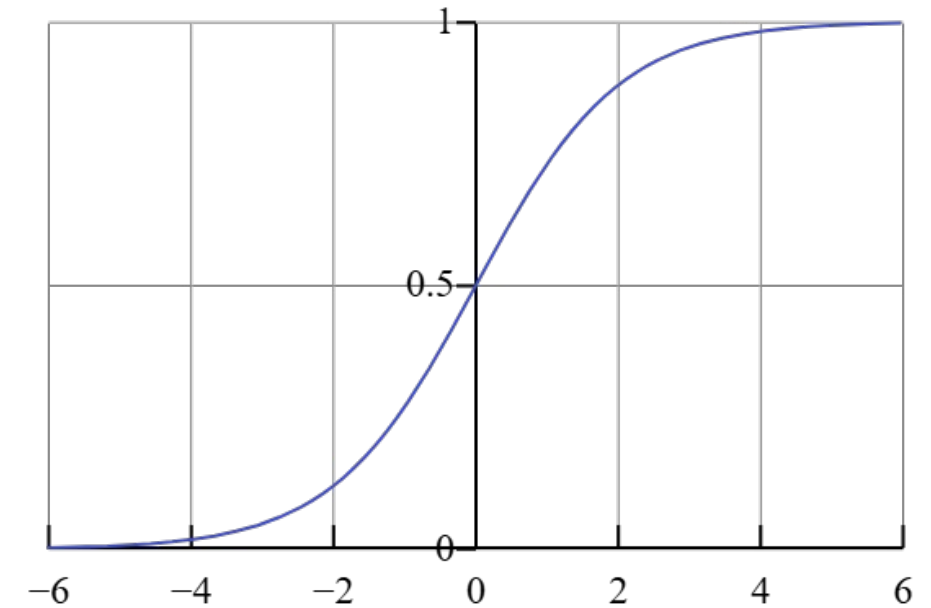
- Clinician: identify clinical pain point or question
- Data engineer: source and clean data
- Data scientist: perform analysis on data, including running models
- Software engineer: build consistent applications that house models
- Clinical Ethicist: Act as consultant regarding bias/safety of model

The Research to Deployment Timeline

- Pose a relevant clinical question
- Software/data team develops possible intervention
- Intervention is compared to humans (or humans +/-) compared (often RCT)
- Assess for clinical safety, efficiency
- Software team begins scaling and deployment
- Post deployment monitoring

Challenges With Clinical Efficiency

- Google's Retinal AI
- What were some challenges to implementation they faced?

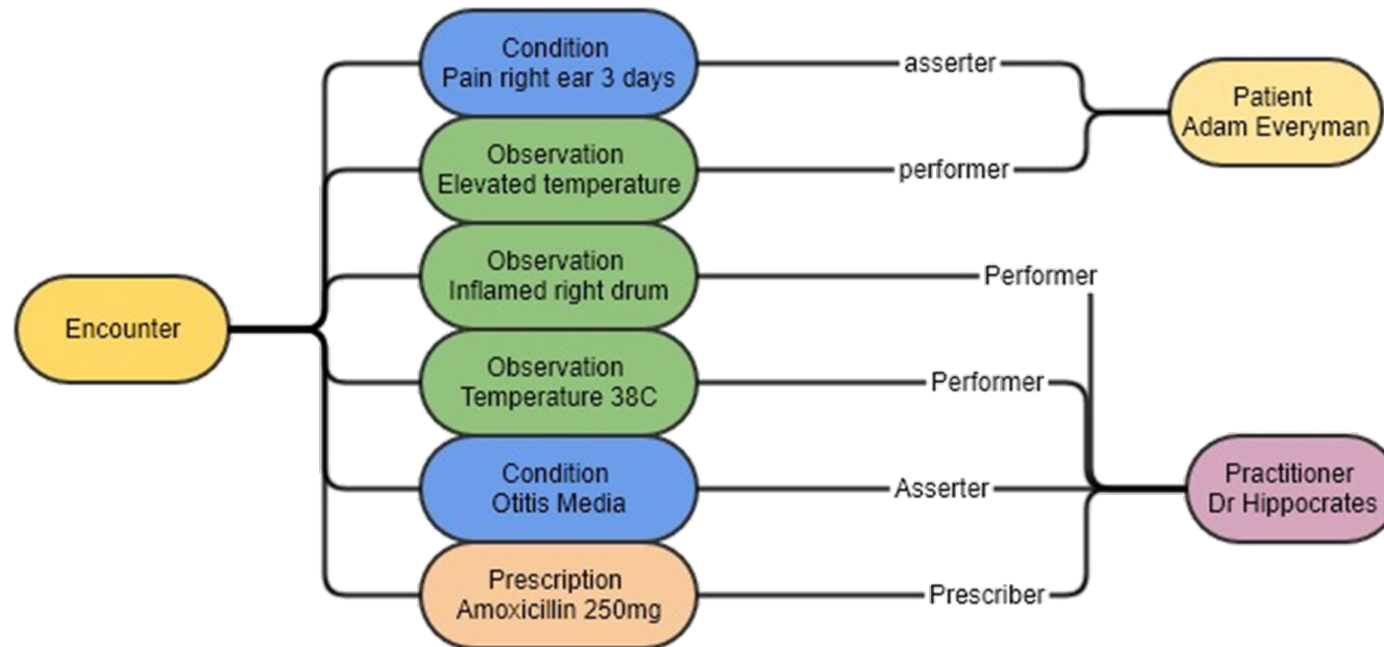


Overview of Clinical Data Ecosystems

- Electronic Health Record (EHR) Systems
- Picture Archiving and Communication Systems (PACS)
- Surgical Robotics Integration

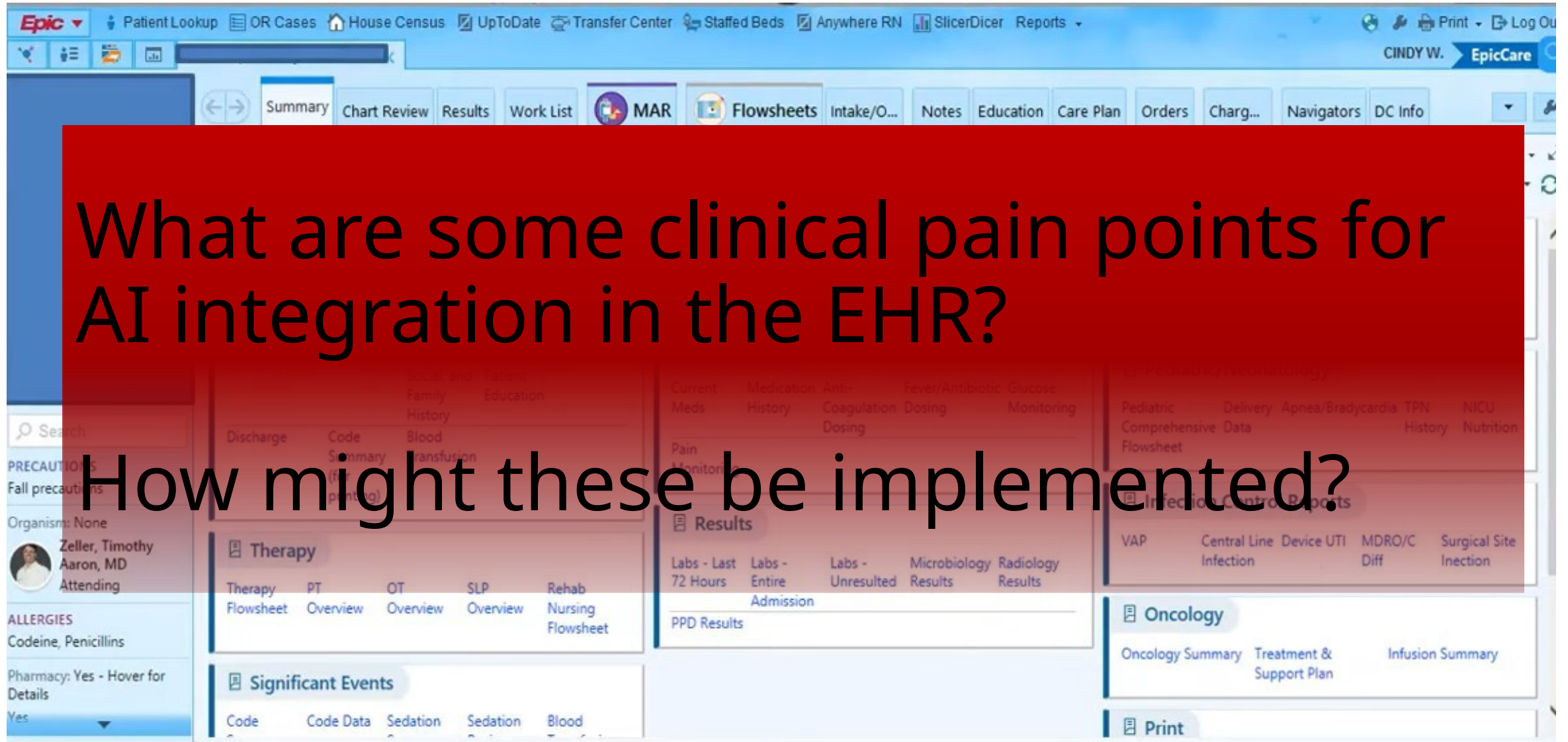
EHR Systems

- Many on the market- they all have their own nuances
- FHIR (Fast Healthcare Interoperability Resources)
- Patient, observation, observer data structure



EHR Integration Points

- Prediction
- Documentation
- Medical Coding
- Operational Optimization (scheduling, billing)



EPIC Sepsis Model

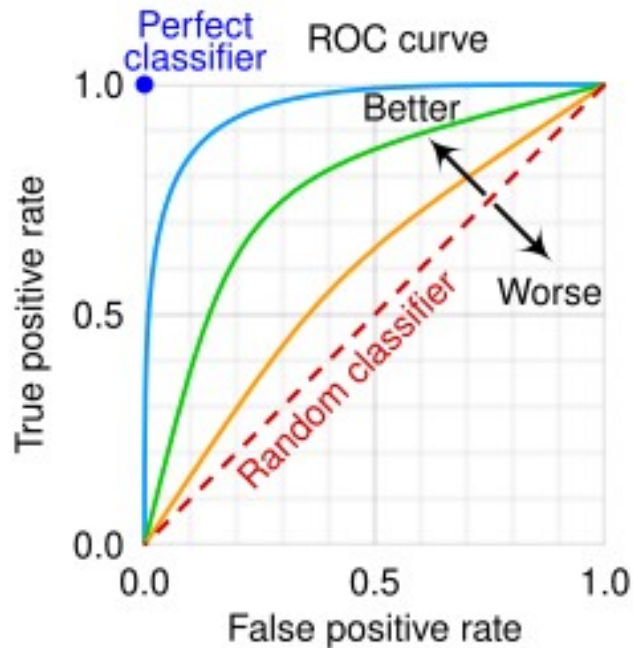
- Model developed by EPIC and integrated into EHR system

Importance The Epic Sepsis Model (ESM), a proprietary sepsis prediction model, is implemented at hundreds of US hospitals. The ESM's ability to identify patients with sepsis has not been adequately evaluated despite widespread use.

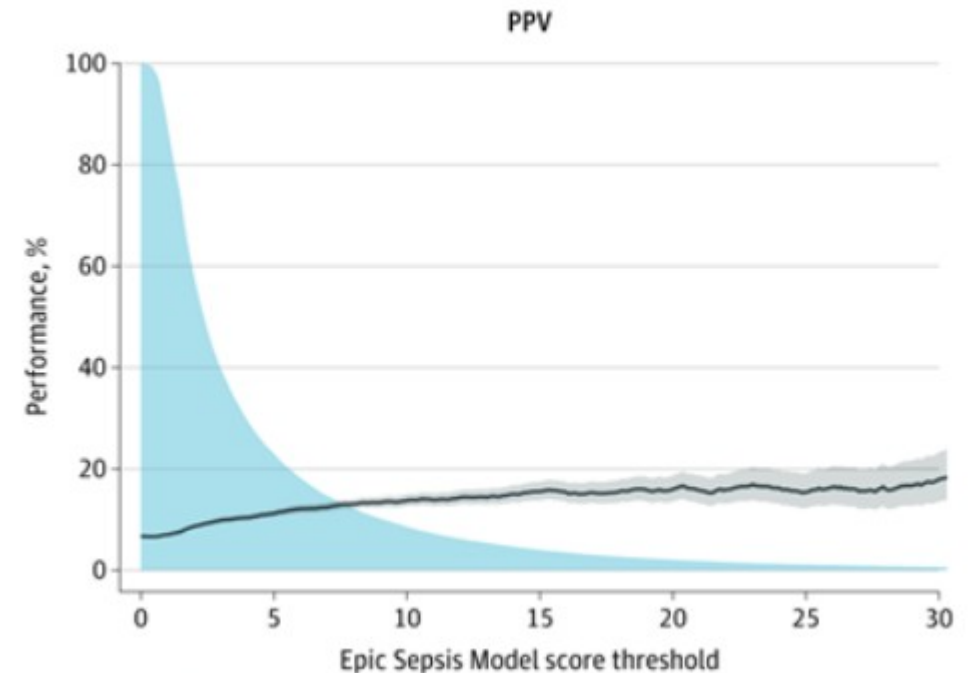
- Per EPIC:
 - Before: SIRS Criteria was flagged, but too many false +
 - Replaced with 60 factor model for assignment of risk score
 - Highlights contribution of features into score

EPIC Sepsis Model

Findings In this cohort study of 27 697 patients undergoing 38 455 hospitalizations, sepsis occurred in 7% of the hospitalizations. The Epic Sepsis Model predicted the onset of sepsis with an area under the curve of 0.63, which is substantially worse than the performance reported by its developer.



Which one corresponds to abo
AUROC = .63



General Problem Statement

Technical Problem Statement

Data

Loss

Network

Training

EPIC Sepsis Takeaways

- Poor PPV -> Alert fatigue, not solving problem it set out to
- Proprietary models need scrutiny d/t conflict of interest
- Let's rate the following
 - Clinical motivation
 - Technical feasibility
 - Clinical utility
- How would we improve this model? (or any model worse than clinicians)

PACS

- Picture Archiving and Communication Systems (PACS)
- Database where Radiology images are stored
- Interface for loading, viewing and reporting on Radiology studies



McKesson PACS

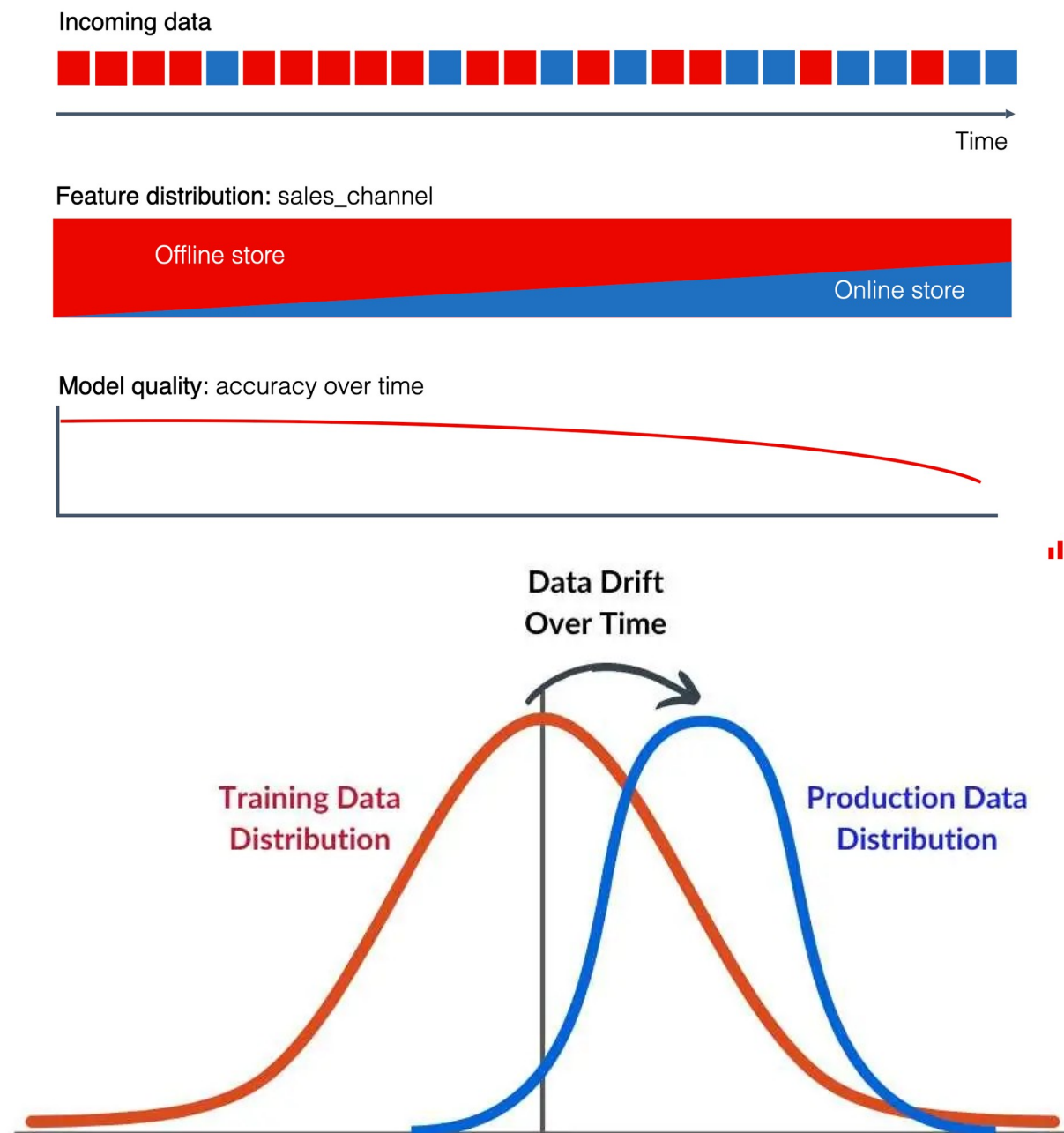
Integration Points

- Prediction
- Documentation
- Medical Coding
- Operational Optimization (scheduling, billing)



Data Drift

- Gradual change over time between training dataset and real world population
- Must consistently monitor algorithmic accuracy compared to ground truth



Procedures

AI-POWERED COLONOSCOPY

NOW AVAILABLE AT
SUNWAY MEDICAL CENTRE,
SUNWAY CITY

Next-Level Colorectal Cancer Detection



Improved
polyp detection
rates



High sensitivity
in distinguishing
precancerous polyps



Enhanced overall
precision of colon
evaluation

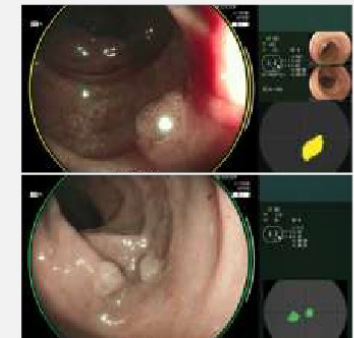


No added risk
to colonoscopy



Non-AI Colonoscopy Image

* Manual detection of cancerous polyps that may require more time



AI-Powered Colonoscopy Image

* Unique Characterisation Mode can instantly detect cancerous polyps

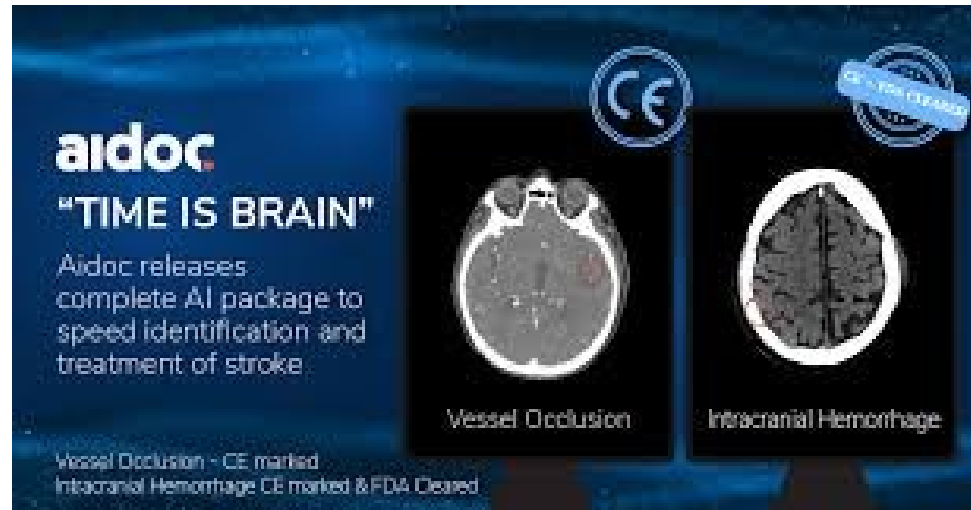
Pathways for Medical Device Approval

- 510k
- De Novo Approval
- Premarket Approval
- Exemptions (rare disease, breakthrough)

For the purposes of the FDA, software (including AI) is considered a medical device

510(k) Pathway for Medical Devices

- FDA approval process for medical devices
- Must find a “predicate” device with same use



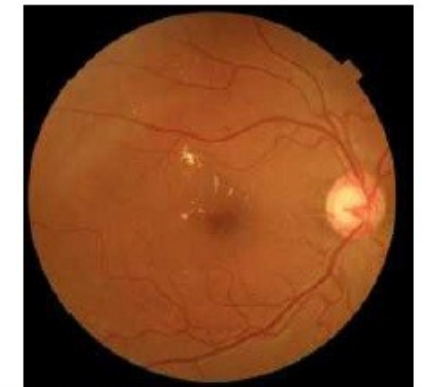
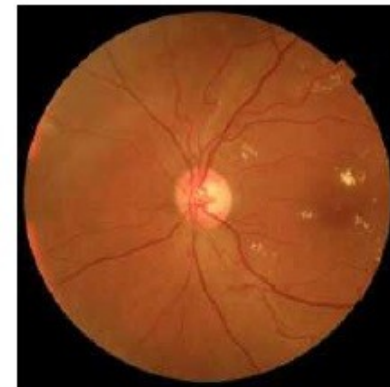
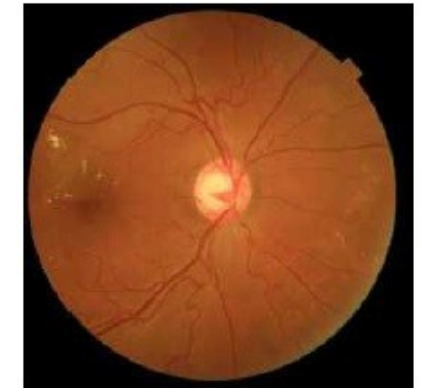
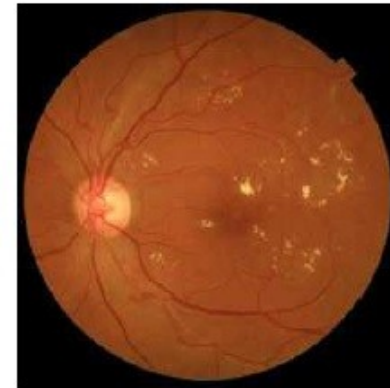
Addressing Data Drift

- Predetermined Change Control Plans (PCCPs)
- Plan outlining what changes and surveillance will occur ahead of time
- Not required for FDA approval

De Novo Device Appro

- For devices that have no precedent
- Ex. IDX: Diabetic retinopathy
- Once approved, can serve as precedent in future 510k

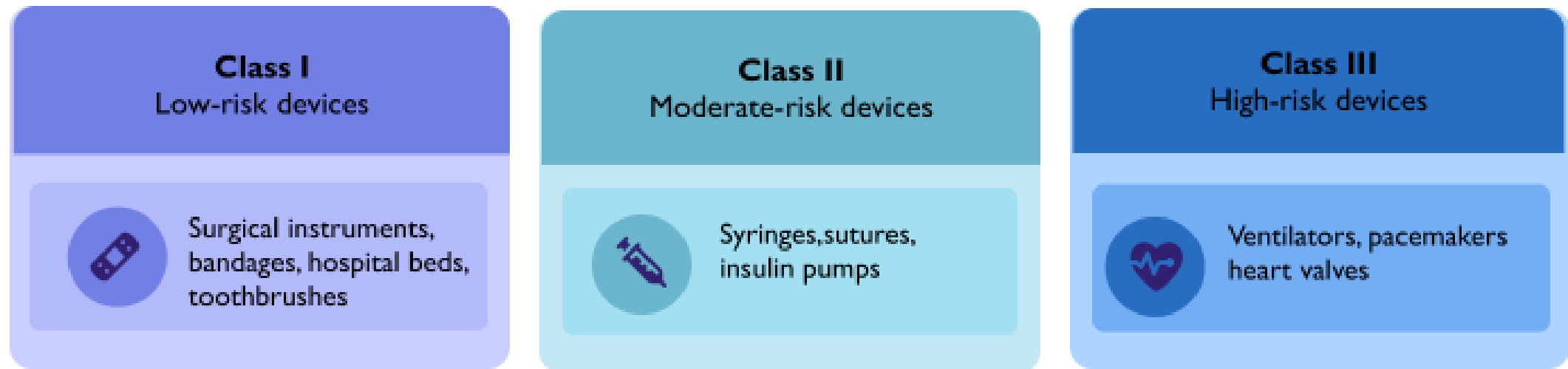
Patient ID:	VTDR
IDx Submission ID:	22-30
Exam Analysis Date:	2017-04-06
Exam Analysis Time:	11:09:15 PM
Exam Result:	Vision-threatening diabetic retinopathy detected



FDA Device Classification

- Class I: Low risk (e.g., stethoscope) – usually exempt.
- Class II: Moderate risk – majority of AI devices fall here.
- Class III: High risk – life-supporting or life-sustaining, needs

PMMA



ARTEREX

FDA Device Classification

- Class I: 510k eligible
- Class II: 510k eligible
- Class III: Needs pre-market approval

Pre Market Authorization

- Needed for high risk medical devices
- Design a study for use
- During the study, obtain an investigational device exemption for the experimental device
- Then, FDA reviews data for clinical approval

Next time:

- Project Intro
- Practice by coming up with a clinical question of your own

Thanks!

Knowledge Distillation

- Used in cases where labelled data is scarce
 - Rare disease
 - Image segmentation

